4 e, f and g represent integers such that

$e \quad f \quad 7 \quad g \quad 12 \quad 17$

is a list of integers written in order of size.

The integers have

a range of 15
a median of 8.5
a mean of 9

Work out the value of e, the value of f and the value of g

range

$17 - e = 15$

$e = 17 - 15 = \underline{\underline{2}}$

mean:

$\cancel{\frac{2 + f + 10 + 12 + 17}{6}} =$

$6 \times \frac{2 + f + 7 + 10 + 12 + 17}{6} = 9 \times 6$

$f + 48 = 54$

$f = 54 - 48$

$\underline{\underline{f = 6}}$

median:

$2 \times \frac{7+9}{2} = 8 \cdot 5 \times 2$

$7 + 9 = 17$

$9 = 17 - 7 = \underline{\underline{10}}$

$e =$ 2

$f =$ 10

$g =$ 6

(Total for Question 4 is 3 marks)

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

$\frac{3 \times 26}{3 \times 7} + \frac{5 \times 7}{3 \times 7}$

$\frac{78}{21} + \frac{35}{21}$

$\cancel{\frac{135}{21}}$ $\frac{113}{21}$

$\therefore \frac{113}{21} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

DO NOT WRITE IN THIS AREA

P 7 9 7 8 5 A 0 5 2 8

Turn over